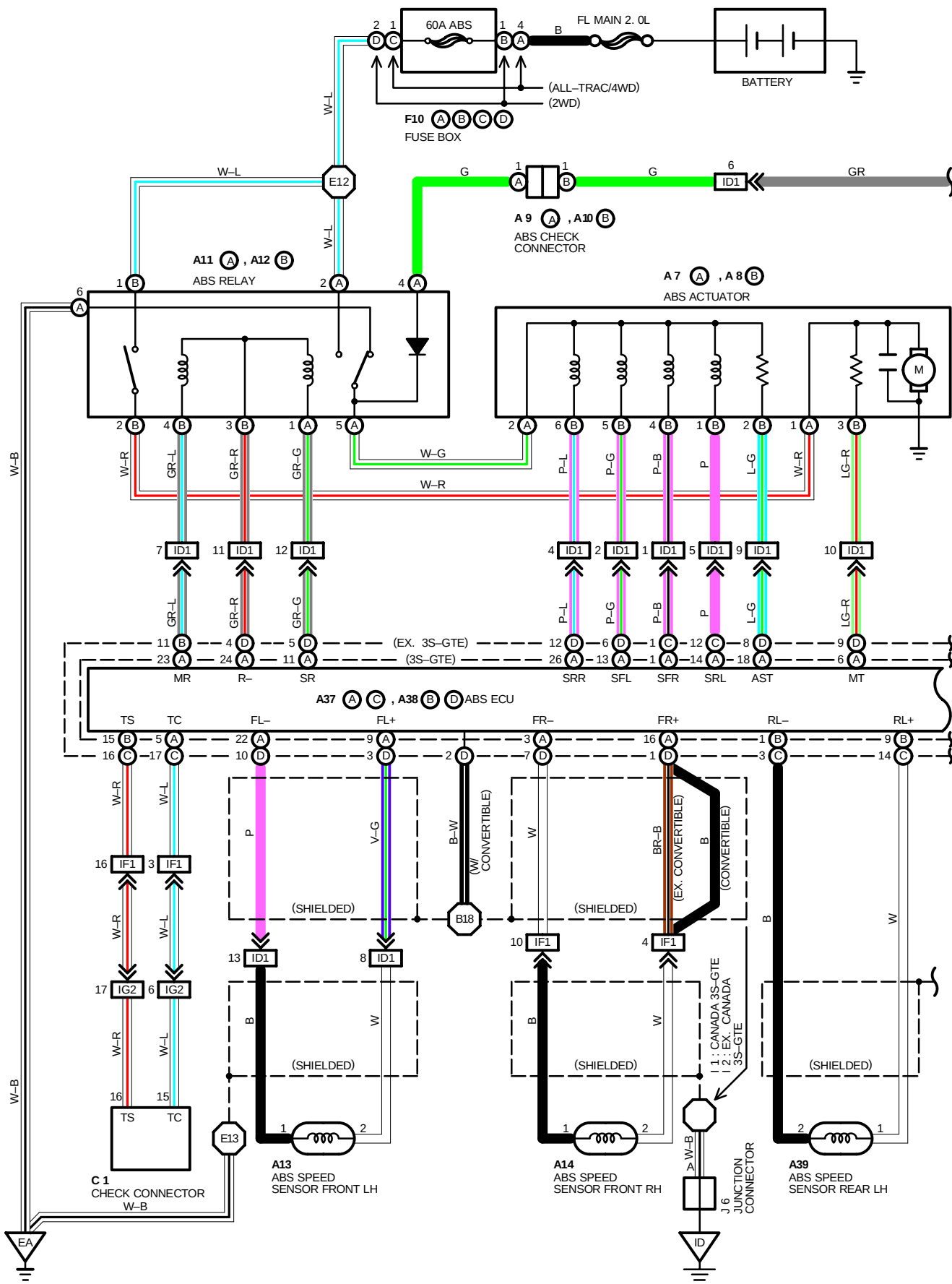


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ABS (ANTI-LOCK BRAKE SYSTEM)

SYSTEM OUTLINE

THIS SYSTEM CONTROLS THE RESPECTIVE BRAKE FLUID PRESSURES ACTING ON THE DISC BRAKE CYLINDERS OF THE RIGHT FRONT WHEEL, LEFT FRONT WHEEL AND REAR WHEELS WHEN THE BRAKES ARE APPLIED IN A PANIC STOP SO THAT THE WHEELS DO NOT LOCK. THIS RESULTS IN IMPROVED DIRECTIONAL STABILITY AND STEERABILITY DURING PANIC BRAKING.

1. INPUT SIGNALS

- (1) SPEED SENSOR SIGNAL
THE SPEED OF THE WHEELS IS DETECTED AND INPUT TO **TERMINALS FL+, FR+, RL+ AND RR+** OF THE ABS ECU.
- (2) STOP LIGHT SW SIGNAL
A SIGNAL IS INPUT TO **TERMINAL STP** OF THE ABS ECU WHEN BRAKE PEDAL IS OPERATED.
- (3) PARKING BRAKE SW SIGNAL
A SIGNAL IS INPUT TO **TERMINAL PKB** OF THE ABS ECU WHEN THE PARKING BRAKE IS OPERATED.
- (4) DECELERATION SENSOR SIGNAL (ALL-TRAC/4WD)
THE DEGREE OF VEHICLE DECELERATION IS DETECTED AND INPUT TO THE ABS ECU.

2. SYSTEM OPERATION

DURING SUDDEN BRAKING THE ABS ECU WHICH HAS SIGNALS INPUT FROM EACH SENSOR, CONTROLS THE CURRENT FLOWING TO THE SOLENOID INSIDE THE ACTUATOR AND LETS THE HYDRAULIC PRESSURE ACTING ON EACH WHEEL CYLINDER ESCAPE TO THE RESERVOIR. THE PUMP INSIDE THE ACTUATOR IS ALSO OPERATING AT THIS TIME AND IT RETURNS THE BRAKE FLUID FROM THE RESERVOIR TO THE MASTER CYLINDER. THUS PREVENTING LOCKING OF THE VEHICLE WHEELS.

IF THE ECU JUDGES THAT THE HYDRAULIC PRESSURE ACTING ON THE WHEEL CYLINDER IS INSUFFICIENT, THE CURRENT ACTING ON THE SOLENOID IS CONTROLLED AND THE HYDRAULIC PRESSURE IS INCREASED. HOLDING OF THE HYDRAULIC PRESSURE IS ALSO CONTROLLED BY THE ECU, BY THE SAME METHOD AS ABOVE. BY REPEATED PRESSURE. REDUCTION, HOLDING AND INCREASE ARE REPLATED TO MAINTAIN VEHICLE STABILITY AND TO IMPROVE STEERABILITY DURING SUDDEN BRAKING.

SERVICE HINTS

A37(A)(C) A38(B)(D) ABS ECU

(CONNECT THE ECU CONNECTOR)

- (B)15, (C)6-GROUND : APPROX. 12 VOLTS WITH THE IGNITION SW AT **ON** POSITION AND CHECK CONNECTOR TS-E1 CONNECTED
- (A) 5, (C)17-GROUND : APPROX. 12 VOLTS WITH THE IGNITION SW AT **ON** POSITION AND CHECK CONNECTOR TS-E1 CONNECTED
- (A)13, (D) 6-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A)11, (D) 5-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A)14, (C)12-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A)18, (D) 8-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A)26, (D)12-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (B)13, (C)10-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION AND ABS WARNING LIGHT GOES OFF
- (A) 2, (C) 2-GROUND : ALWAYS CONTINUITY
- (A)15, (C)12-GROUND : ALWAYS CONTINUITY
- (A)12, (C)11-GROUND : APPROX. 12 VOLTS WITH IGNITION SW AT **ON** POSITION
- (B) 6, (C) 7-GROUND : APPROX. 12 VOLTS WITH BRAKE PEDAL DEPRESSED
- (A)25, (C)22-GROUND : ALWAYS APPROX. 12 VOLTS
- (B)14, (C)16-GROUND : APPROX. 12 VOLTS WITH ENGINE RUNNING AND PARKING BRAKE LEVER RELEASED

(DISCONNECT THE ECU CONNECTOR)

- (A) 1-(A)18, (C) 1-(D) 8 : APPROX. 6 Ω
- (A)13-(A)18, (D) 6-(D) 8 : APPROX. 6 Ω
- (A)14-(A)18, (C)12-(D) 8 : APPROX. 6 Ω
- (A)26-(A)18, (D)12-(D) 8 : APPROX. 6 Ω
- (A) 9-(A)22, (D) 3-(D)10 : APPROX. 0.8-1.3 Ω
- (A)16-(A) 3, (D) 1-(D) 7 : APPROX. 0.8-1.3 Ω
- (A)11-(A)24, (D) 5-(D) 4 : APPROX. 60-100 Ω
- (A)23-(A)24, (D)11-(D) 4 : APPROX. 50-80 Ω
- (B) 8-(B)16 : APPROX. 0.8-1.5 K Ω
- (C)19-(C)20 : APPROX. 1.1-1.7 K Ω

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
A 7	A	28 (3S-GTE), 30 (5S-FE)	A38	B	34 (L/B), 36 (CONVERTIBLE)
A 8	B	28 (3S-GTE), 30 (5S-FE)		D	34 (L/B), 36 (CONVERTIBLE)
A 9	A	28 (3S-GTE), 30 (5S-FE)	A39		34 (L/B), 36 (CONVERTIBLE)
A10	B	28 (3S-GTE), 30 (5S-FE)	A40		34 (L/B), 36 (CONVERTIBLE)
A11	A	28 (3S-GTE), 30 (5S-FE)	A43		34 (L/B), 36 (CONVERTIBLE)
A12	B	28 (3S-GTE), 30 (5S-FE)	B 2		28 (3S-GTE), 30 (5S-FE)
A13		28 (3S-GTE), 30 (5S-FE)	C 1		28 (3S-GTE), 30 (5S-FE)
A14		28 (3S-GTE), 30 (5S-FE)	C13	B	32
A37	A	34 (L/B), 36 (CONVERTIBLE)	C14	A	32
	C	34 (L/B), 36 (CONVERTIBLE)	F10	A	28 (3S-GTE)

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1B	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
1F	20	FLOOR WIRE AND J/B NO. 1 (LEFT KICK PANEL)
3B	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
ID1	44	ENGINE ROOM MAIN WIRE AND FLOOR WIRE (LEFT KICK PANEL)
IF1	44	FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)
IG2	44	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)

▽ : GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EA	38 (3S-GTE)	FRONT RIGHT FENDER
	40 (5S-FE)	
EB	38 (3S-GTE)	FRONT LEFT FENDER
	40 (5S-FE)	
ID	44	LEFT KICK PANEL
BI	48 (L/B)	UNDER THE LEFT CENTER PILLAR
	50 (C/P)	
	52 (CONVERTIBLE)	

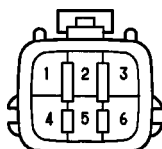
○ : SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E12	38 (3S-GTE)	ENGINE ROOM MAIN WIRE	B 1	50 (C/P)	FLOOR WIRE
	40 (5S-FE)			52 (CONVERTIBLE)	
E13	38 (3S-GTE)		B 7	48 (L/B)	
	40 (5S-FE)		B15	48 (L/B)	
I 1	46	COWL WIRE	B18	50 (C/P)	
I 2				52 (CONVERTIBLE)	
B1	48 (L/B)	FLOOR WIRE			

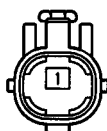
A 7 (A) GRAY



A 8 (B) GRAY



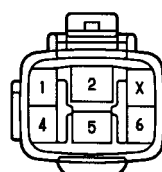
A 9 (A) DARK GRAY



A10 (B) DARK GRAY



A11 (A) GRAY



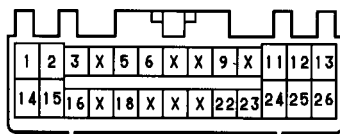
A12 (B) GRAY



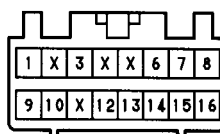
A13, A14 GRAY



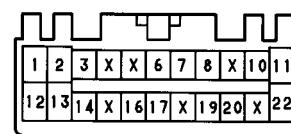
A37 (A) DARK GRAY



A38 (B) DARK GRAY



A37 (C) DARK GRAY



ABS (ANTI-LOCK BRAKE SYSTEM)

